STATEMENT OF WORK

PBA Testing, Lab Services and Engineering

I. TITLE: Testing, Lab Services and Engineering

II. BACKGROUND: Pine Bluff Arsenal (PBA) performs a variety of missions for the Department of Defense, including the manufacture and demilitarization of ammunition and related components. To accomplish this mission within the environmental protections established by the Clean Air Act (CAA), Clean Water Act (CWA), Resource Conservation and Recovery Act (RCRA), the Comprehensive Environmental Restoration and Compensation Liability Act (CERCLA), and the Spill Prevention, Control, and Countermeasure (SPCC) regulations. In addition to the aforementioned regulations, to accomplish the mission, PBA must comply within requirements established by the Defense Environmental Restoration Program (DERP). PBA is seeking contract services related to these requirements. This effort will augment on-going on-site laboratory evaluations in several media. An existing contract that provides these services will expire in December 2026. All references to "PBA" imply action, coordination and/or approval of the Contracting Officer (KO).

This contract will task the contractor with:

Task 1 Ecological Surveys;
Task 2 Toxicity Testing Compliance required by permit;
Task 3 Groundwater Assessments;
Task 4 Geographic Information System (GIS) Support;
Task 5 Stormwater Compliance;
Task 6 Regulatory Compliance Assistance;
Task 7 Restoration Regulatory Support
Task 8 Domestic Water Supply Well Removal
Task 9 Travel reimbursement.

The contract will be for an initial base year with four renewable option years for a possible total of five years. The contractor shall provide on-call assistance to answer technical questions concerning the contract deliverables for the duration of the contract. Workdays for this contract are Monday through Friday beginning at 7:00 AM and ending at 5:00 PM, Central Time zone. Five (5) day advance notification of contractor site visits is required. All contractor personnel performing activities at PBA must attend a Safety Briefing before any work is conducted. The safety Briefing is held Monday-Thursday beginning at 7:30 AM at Building 17-130.

The local work covered by this contract is to be performed at PBA. Postal address is Pine Bluff Arsenal, 10020 Kabrich Circle, Pine Bluff, Arkansas 71602-9500. The Arsenal is located off Highway 530, Exit 32, about 12 miles northwest of Pine Bluff and 35 miles southeast of Little Rock. Note: The contractor shall include the contract number, the COR's name, and the COR's phone number on all items shipped to PBA.

To perform services in this Scope of Work certain accreditations and professional certification are required. All laboratories analyzing water, sediment, soil and conducting toxicity testing for this contract shall have a current Environmental Laboratory Accreditation from Arkansas Energy and Environment. Additionally, professionals preparing and reviewing SPCC Plan, RCRA, and CWA

permits shall have a professional engineering licensure in Arkansas. Professionals preparing and reviewing groundwater reports shall have a professional geologist licensure in Arkansas.

Limited Emergency first aid, to prevent loss of life or limb, is available for contractor personnel at the U.S. Army Health Clinic at PBA for injuries sustained in the performance of their duties. This service is normally available from 0630 to 1630, Monday through Thursday. For any other injury, contractor personnel should go to Jefferson Regional Medical Center or see their private physician. The requirements of Department of Labor, Occupational Safety and Health Administration (OSHA) Standard 29 CFR 1910 will be followed.

All contractor and sub-contractor personnel will adhere to all applicable PBA and Army regulations.

No fire or flame-producing device will be used on the Arsenal without prior approval of the Contracting Officer. The contractor shall also be required to obtain a burn or hot work permit from the Fire Department.

Smoking will not be permitted, except in designated areas.

Equipment will be maintained in a safe operating condition and a fire safe condition at all times. It shall be stored in a designated area, approved by the Contracting Officer.

Hoists, cranes, derricks, or similar equipment shall not be operated where it is possible to bring any part of the machine, suspended load, or lines closer than ten feet to power lines unless the lines have been de-energized and grounded.

III. TASKS

Task Number 001: Ecological Surveys

Ecological Surveys:

Task 1.1 The contractor shall conduct Ecological Surveys as required by the Division of Environmental Quality (DEQ) and in the RCRA and NPDES permits. These surveys will assess the current water, sediment and biological status of the Arsenal's surface waters. Additionally, these surveys will determine the effectiveness of corrective measures, long-term pollution abatement measures and any other required regulatory compliance measures. On an annual basis, the contractor shall conduct ecological monitoring and assessment of the Arsenal's Solid Waste Management Unit (SWMU) closure sites following the 2008 Ecological Monitoring Plan[1] (GLEC 2008). The monitoring will measure the selected contaminants and toxicity from samples collected from surface waters, sediments and fish tissues at the following four areas of the PBA site:

1. SWMU 75. To address surface water and sediment quality within Phillips Creek, in particular both up- and down-stream of the former remnant DDT source area removal that was completed during 2004. The following will be sampled in Phillips Creek: Five (5) surface water samples, 5 sediment samples, and fish tissue samples at 5 locations.

2. SWMUs 23 and 24. To address sediment quality downstream of the soil removal at SWMU 24, and soil removal and capping activities at SWMU 23 that were conducted during 2004. Ecological monitoring will be conducted at up- and down-gradient locations in a tributary leading to the RCI. The following will be sampled: Three (3) surface water samples, 3 sediment samples, 3 samples for sediment toxicity testing.
3. SWMUs 27 and 28. To address surface water and sediment quality near the Solid Waste ARKLA Site (SWMU 27) and the Old ARKLA Rubble Site (SWMU 28). Ecological monitoring will be conducted at several up- and down-gradient locations. The following will be sampled: Six (6) surface water samples, and 6 sediment samples, 6 samples for sediment toxicity testing.
4. SWMUs 53, 55, 60. To address surface water and soil/sediment quality near the South Depot Burning Pit (SWMU 53), the Current Sanitary Landfill (SWMU 55), and White Phosphorus Slab Burn Disposal Area (SWMU 60). Ecological monitoring will be conducted at several up- and down-gradient locations. The following will be sampled: Six (6) surface water samples, and 6 sediment/soil samples, 6 samples for sediment/soil toxicity testing.

Sample sites will consist of up-gradient sites to serve as an indicator of background conditions and down-gradient sites to determine if the contaminants are entering the surface water systems.

Task 1.2 Results of the ecological monitoring and assessment will be reported in the Annual Groundwater Assessment and Ecological Trend Monitoring report. The Annual report will include a description of the sampling, analysis and comparisons to background and approved ecological screening levels that are contained in the Remedial Action Decision Document (RADD; DEQ, 2009). This report will be provided to PBA Environmental in draft format for review prior to being submitted to DEQ; the Annual report is completed and submitted to DEQ during September through December of the following year.

Task 1.3 Once every 5 years beginning with 2028 the contractor shall prepare a 5-year Groundwater Assessment and Ecological Trend Monitoring report. The contractor shall evaluate trends in surface water and sediment quality data; the next 5-year report will address reporting years 2024-2029 and will be submitted to DEQ during September through December of 2029. The 5-year report replaces the Annual report with respect to the 2029 ecological monitoring data.

Task 001 References (Available on request):

1. GLEC. 2008. Groundwater and Ecological Monitoring Plans, Volume 4 Corrective Measures Study, Pine Bluff Arsenal, Pine Bluff Arkansas. Great Lakes Environmental Center, Columbus, Ohio. Report date: Draft May 2004, Final: November 2008.
2. PBA. EES Additional SOW for Continues support for Mercury Criterion development, CAO Support, and Renewal of NPDES Permit.
3. PBA. EES Additional SOW for CAO support and Renewal of NPDES Permits.
4. PBA. SOW for Additional Ecological Survey: Bioaccumulation Factor Mercury Study Execution.

Task Number 2: Toxicity Testing Compliance

The contractor shall provide off-site services in support of this task.

Task 2.1 – Acute and Chronic Bioassays (four outfalls)

This subtask is for toxicity testing at four PBA NPDES outfalls (008, 010, 012, 01C) in NPDES Permit AR0001678. The acute effects of PBA effluents on the survival of *Daphnia pulex* and fathead minnows will be evaluated using 48-hour static-renewal toxicity tests at three outfalls (008, 010, 012). The chronic effects of PBA effluents on *Ceriodaphnia* and fathead minnows will be evaluated using 7-day static-renewal toxicity tests at one outfall (01C). The toxicity testing program for the effluent samples will be conducted according to the specific regulatory requirements contained in NPDES Permit AR0001678. Currently, the permit requires quarterly testing at four outfalls.

Task 2.2 - The contractor shall provide additional support for toxicity testing during the renewal of the NPDES permit, response to comments or additional requirements identified by DEQ.

OPTION 2.1 - Toxicity Identification/Reduction Evaluations (one outfall)

If there is unacceptable toxicity in one or more of PBA's NPDES effluents, this option will be utilized to identify, by characterization, fractionization, toxicity testing and chemical analyses, the components of effluents responsible for its toxicity and determine the effectiveness of pollution control options. This effort will consist of engineering and chemistry support services to correlate activity with toxicity and to perform engineering evaluations to reduce the hazard of PBA discharges to receiving streams.
Frequency: Potentially two (2) evaluations within the contract POP (Base-OY4)

Task 002 References (Available on request):

1. PBA. EES Additional SOW for Continued support for Mercury Criterion development, CAO Support, and Renewal of NPDES Permit.
2. PBA. EES Additional SOW for CAO support and Renewal of NPDES Permits.

<u>Task Number 3: Groundwater Hydrogeological Assessments</u>

The contractor shall provide support to conduct groundwater hydrogeological assessments that involve the annual statistical assessment of groundwater analyses data from three RCRA Permit sites and 12 RCRA Corrective Action sites located on Pine Bluff Arsenal. The groundwater hydrogeological assessments incorporate groundwater data from 111 monitoring wells that are included in Pine Bluff Arsenal's groundwater monitoring program. DEQ requires that these assessments be performed by an Arkansas Certified Geologist.

The contractor shall prepare two types of RCRA groundwater assessment reports annually, these include:

Task 3.1: **Detection Monitoring of Runoff Control Impoundment (RCI) and 83 Hazardous Waste Landfill**

Detection Monitoring groundwater assessment reports for 2 Regulated Units subject to RCRA Permit 1H-RN3[1]. Annually, one separate report for each of the following:

a. Runoff Control Impoundment (RCI)
b. 83 Hazardous Waste Landfill

The RCRA Detection Groundwater Assessment reports for the RCI and 83 HWL are provided to DEQ by May 1st of the following year.

The RCRA Detection Monitoring program that is specified in Permit 1H-RN3 requires a statistical evaluation of intrawell trends in concentration to determine if a statistically significant increase has occurred at the site that would be indicative of a release. As specified in the Permit 1H-RN3, nonparametric and parametric statistical procedures are used to identify outliers and to evaluate trends in the concentrations of the monitored parameters.

Two procedures are utilized to evaluate the groundwater quality data. For compounds that are detected in 80% or more of the groundwater samples, Shewhart-CUSUM control charts are constructed. For all other compounds, a non-parametric prediction limit is utilized that is set at twice the required practical quantification limit (RPQL) listed in Table 4.1a of Pine Bluff Arsenal's Quality Assurance Project Plan. The preparation of Shewhart-CUSUM Control charts involve:

1. Determination if the data are normally or log normally distributed using either the Shapiro-Wilks Test for Normality ($n<50$) or Shapiro-Francia Test for Normality ($n>50$).
2. Evaluation of the data set for outliers using either Dixon's Test (for less than 25 data points) or Rosner's Test for Outliers (for more than 25 data points), outliers are removed from the data set and subsequent statistical analysis.
3. If outliers are removed, the Shapiro-Wilks (or Shapiro–Francia) Test for Normality is updated.
4. Preparation of a Mann-Kendall Trend Analysis to determine if there is an upward trend in the data since groundwater monitoring was initiated.
5. Preparation of a Shewhart-CUSUM Control Chart.
6. Preparation of a time series graph.

With respect to the RCRA Detection Monitoring groundwater assessment reports, each parameter for each well for each of the sampling rounds is evaluated using either Shewhart-CUSUM control charts or non-parametric prediction limits.

Task 3.2 – Detection Monitoring of 86 Hazardous Waste Landfill

Detection Monitoring groundwater assessment reports one Regulated Units subject to RCRA Permit 1H-RN3[1]. Annually, one separate report for each of the following (three reports total):

a. 86 Hazardous Waste Landfill

The RCRA Detection Groundwater Assessment reports for the 86 HWL are provided to DEQ by May 1st of the following year.

The RCRA Detection Monitoring program that is specified in Permit 1H-RN3 requires a statistical evaluation of intrawell trends in concentration to determine if a statistically significant increase has occurred at the site that would be indicative of a release. As specified in the Permit 1H-RN3, nonparametric and parametric statistical procedures are used to identify outliers and to evaluate trends in the concentrations of the monitored parameters.

Two procedures are utilized to evaluate the groundwater quality data. For compounds that are detected in 80% or more of the groundwater samples, Shewhart-CUSUM control charts are constructed. For all other compounds, a non-parametric prediction limit is utilized that is set at twice the required practical quantification limit (RPQL) listed in Table 4.1a of Pine Bluff Arsenal's Quality Assurance Project Plan. The preparation of Shewhart-CUSUM Control charts involve:

1. Determination if the data are normally or log normally distributed using either the Shapiro-Wilks Test for Normality ($n<50$) or Shapiro-Francia Test for Normality ($n>50$).
2. Evaluation of the data set for outliers using either Dixon's Test (for less than 25 data points) or Rosner's Test for Outliers (for more than 25 data points), outliers are removed from the data set and subsequent statistical analysis.
3. If outliers are removed, the Shapiro-Wilks (or Shapiro–Francia) Test for Normality is updated.
4. Preparation of a Mann-Kendall Trend Analysis to determine if there is an upward trend in the data since groundwater monitoring was initiated.
5. Preparation of a Shewhart-CUSUM Control Chart.
6. Preparation of a time series graph.

With respect to the RCRA Detection Monitoring groundwater assessment reports, each parameter for each well for each of the sampling rounds is evaluated using either Shewhart-CUSUM control charts or non-parametric prediction limits.

Task 3.3: **RCRA Corrective Action Sites**
Annual groundwater assessment report to address groundwater quality impacts at 12 RCRA Corrective Action sites (Closure Sites). Annually, one report (includes ecological monitoring, See Task 1.2) is prepared and includes the assessment of each of the following sites (one report).

a. 504th Street Burn Site (Site 4a; SWMU 25);
b. Lewisite and Mustard Agent Areas (Site 7; SWMUs 19, 20, 21, 22);
c. North HC Smoke Test Pond (Site 8; SWMU 74);
d. Former Chemical Manufacturing Area (Site 11; SWMUs 18, 23, 24, 29, 75);
e. Open Burn/Open Detonation Pad (Site 12; SWMU 7);
f. South Depot Burning Pit (Site 20a; SWMU 53);
g. HC Test Pond (Site 23a; SWMU 62);
h. Thermite Disposal Area (Site 24; SWMU 63);
i. Agent BZ Pond (Site 27; SWMU 65);
j. Solid Waste ARKLA Site (Site 29; SWMU 27);
k. Bombing Mat Vicinity (Site 62, SWMU 80); and
l. Yellow Lake Borrow Area (Site 63, SWMU 81).

The groundwater assessments for each of the 12 sites are included in the Annual Groundwater Assessment and Ecological Trend Monitoring report and is submitted to DEQ between September and December of the following year.

Each report includes:

1. Evaluation of the rate of groundwater flow and preparation of groundwater contour maps;
2. Tables summarizing groundwater quality data since initiation of Corrective Action groundwater monitoring during 2004;
3. A Level 2 laboratory data package reports for each semi-annual sampling event; and
4. Evaluation of quality control samples (field blanks, duplicates, quality control sample).

Task 3.4: In addition, a 5-year Groundwater Assessment and Ecological Trend Monitoring report is prepared to evaluate trends in groundwater quality data for each of the 12 sites subject to RCRA Corrective Action groundwater monitoring; the next 5-year report will address reporting years 2022-2026 and will be submitted to DEQ during September through December of 2027. The 5-year report replaces the Annual report with respect to the 2026 groundwater monitoring data.

Task 3 References (Available on request):

1. DEQ. 2018. Final RCRA Hazardous Waste Permit 1H-RN3.

Task Number 4: Geographic Information System (GIS) Support

The contractor shall provide ongoing maintenance and updates to Pine Bluff Arsenal's GIS, including: (1) maintaining the GIS database and associated data layers; (2) periodically uploading groundwater analytical data provided or approved by Pine Bluff Arsenal or USACE; (3) updating GIS maps with current and historical aerial imagery and other relevant data furnished by Pine Bluff Arsenal or USACE; and (4) providing training to designated Pine Bluff Arsenal personnel on the operation and basic maintenance of the GIS database. The contractor shall coordinate data updates and map revisions with Pine Bluff Arsenal and USACE, as applicable.

Outsourcing this effort is justified because Pine Bluff Arsenal currently lacks sufficient personnel, specialized training, technical expertise, software, hardware, and other tools necessary to perform these GIS maintenance and data-management services in-house. Contractor support is therefore required to maintain the GIS in a timely, accurate, secure, and technically appropriate manner. The contractor shall coordinate all data updates and map revisions with Pine Bluff Arsenal and USACE, as applicable, and shall perform only those activities authorized by the Government. An RSCA for these services has been found acceptable and approved.

This effort is a separate CLIN and shall not be included in the DERA Program Management CLIN.

Task Number 5: Stormwater Compliance

Task 5.1 The contractor will provide services for the sampling of selected stormwater sites located on the Arsenal. Stormwater collection will be at three stormwater outfalls as required by PBA's Stormwater Discharge Permit, ARR00A048, specifically Outfalls 4, 8 and 10. All stormwater sampling and testing will be in accordance with PBA's NPDES stormwater permit and the PBA Stormwater Pollution Prevention Plan (SWPPP). All collected samples with respective chains of custody will be delivered to a contract laboratory certified by DEQ for analysis of the parameters for total suspended solids and chemical oxygen demand. The contractor shall submit to PBA the laboratory analysis containing all the pertinent data from field tests and if necessary, a Corrective Action Plan within 30-days of sampling. Every effort will be made to obtain the required samples according to the sampling protocols of the permit. The contractor shall also provide a draft of the SWAR (Storm Water Annual Report) on or before January 16 of each year.

Task 5.2 The contractor will provide additional Stormwater support for IGP permit renewals or modification, application preparation, application support, SWPPP revisions, and any site visits as necessary.

Task 5.3 – SWPPP and/or Spill Prevention Training (once/year) – The contractor shall provide training to employees responsible for complying with Industrial General Permit and Construction Stormwater requirements.

Option 5.1 – Construction Stormwater Support – The contractor shall provide assistance with SWPPP development including mapping, project planning and onsite meetings.

Task Number 6: Regulatory Compliance Assistance

The contractor will provide multimedia compliance services. These are recurring and required by the Clean Air Act (CWA), Clean Water Act (CWA), Resource Conservation and Recovery Act (RCRA), and CERCLA.

Task 6.1 - SPCC – The contractor shall provide SPCC Plan review, revisions FY27-FY29. Annual review and revision will be required once per year.

Task 6.2 – RCRA – Permit modification. Contractor shall assume RCRA permit modification twice during the contract period. Support will include application preparation, application support, draft permit review, meetings with DEQ and response to comments (Anticipate once/ 2 years beginning OY2).

Option 6.1 – RCRA – The contractor shall provide services for the renewal of PBA's RCRA Permit 1H-RN3. Services include application preparation, draft permit review, meetings with DEQ and response to comments (as necessary). Frequency: Once every 10 years. (Anticipated FY27).

Option 6.2 – SPCC - SPCC Plan is required to be renewed and recertified every 5 years. The contractor shall assist in preparing and updating the plan consistent with 40 CFR Part 112. Renewal and recertification include plan review and plan updates (as necessary). Frequency: Once every 5 years (Anticipated FY30).

Option 6.3 – CAA - Triennial Emissions Inventory. Contractor shall support the preparation of report and uploading to applicable online database. Reports shall be submitted to PBA, OY1, 2027 and April 1, 2030 (Frequency: Once every 3 years)

Option 6.4 – CAA - Title V Renewal. Contractor shall prepare applications, provide application support, draft permit review, additional model runs and response to comments (as necessary). Frequency: Once every 5 years (Anticipated FY30, OY4)

Option 6.5 – CAA - Title V Regulatory Support. Work to include application preparation, application support, draft permit review, response to comments, additional model runs, assistance with Semi-Annual Monitoring (SAM) reporting and Annual Compliance Certification (ACC). (Anticipate once /year)

Option 6.6 – CWA - Permit Renewal. Work to include application preparation, application support, draft permit review, and response to comments (as necessary). Application must be submitted by July 4, 2027, 6 months prior to expiration date of January 31, 2028. Frequency: Once every 5 years (Anticipated FY27)

Outsourcing this effort is justified because Pine Bluff Arsenal currently lacks sufficient personnel, specialized training, technical expertise, and other tools necessary to perform these services in-house. Contractor support is therefore required to prepare these compliance services in a timely, accurate, secure, and technically appropriate manner. The contractor shall coordinate all updates and responses with Pine Bluff Arsenal and USACE, as applicable, and shall perform only those activities authorized by the Government. An RSCA for these services has been found acceptable and approved.

Task Number 7 – Restoration Regulatory Support

The contractor will provide Defense Environmental Restoration Program multimedia compliance services. Regulatory drivers for services include Division of Environmental Quality CWA and RCRA.

7.1: Interim Effluent Condition (IEC) -Third Party Rulemaking for RCI Outfall 01C

The contractor shall provide administrative law, technical and meeting attendance support services to include an Interim Effluent Condition, equivalent to an interim water quality standard for mercury (Hg) into PBA's NPDES Permit AR0001678, Outfall 01C. The proposed standard must be approved by the Division of Environmental Quality, the Arkansas Pollution Control and Ecology Commission, the Arkansas State Legislature, and the Region 6 Environmental Protection Agency (EPA). Working with the PBA legal team, the contractor shall provide administrative legal and technical support for the following steps of the process:

1. **Petition the Commission –** Pine Bluff Arsenal petitions the Pollution Control and Ecology Commission (PC&EC) to adopt revisions proposed in the supporting technical documentation.
2. **Public Notice –** Rule 8 – Administrative Procedures requires that information about the proposed revisions be published in newspapers with statewide circulation. Copies of the proposed regulation are made available at DEQ headquarters and website.

3. **Public Hearings –**A public hearing is held to gather public input on the proposed revisions. The hearing is held at least 20 days after the date of public notice. During the hearing, written and oral comments concerning the proposed revisions are accepted and entered into the public record.
4. **Responsive Summary –** PBA must read and publicly respond to all comments received during the public comment period. Both DEQ and Responsive Summary are published on DEQ's website.
5. **Update Governor's Office –** If needed, the final draft of Rule 2 – Arkansas Water Quality Standards is sent to the governor's office for review if substantive changes were made based on public comment. [Technical support if needed]
6. **Commission Adoption –** After the response to comments is developed, the final draft of the rule is sent to PC&EC to be adopted. [Providing administrative law and technical support, attendance of onsite meeting required and providing support services to the PBA legal team before the Governor-appointed Commission]
7. **Legislative Approval –** After adoption by the PC&EC, the rule is then sent to two legislative committees for review: Public Health, and Rules and Regulations. [Administrative law and technical support, attendance of onsite meeting required, and providing support before legislative committees]
8. **Inclusion into NPDES Permit** - Once the standard change is approved by Region 6 EPA the contractor shall provide services to incorporate the approved Interim Effluent Condition temporary water quality standard into NPDES AR0001678. Tasks include application preparation, application support, draft permit review, and supportive responses to comments (as necessary).

7.2: RCI Hydraulic Retention Time Improvements – The contractor shall provide support based on the approved Pollution Minimization Plan, included in the Interim Effluent Condition. Tasks include technical and potentially construction support to improvement operational performance and hydraulic retention time of the RCI.

7.3 Philips Creek Impact on Arkansas River

7.3.1 The contractor shall propose the services necessary to assess the flow of the Arkansas River relative to Phillips Creek at their confluence; however, such services shall be performed only upon a determination by the U.S. Army and Pine Bluff Arsenal that they are legally permissible, justified, and authorized.

7.3.2 The contractor shall propose services to assess contaminant trends in fish tissue, surface water, and sediment quality in the Arkansas River and at the Phillips Creek confluence, including downstream transects focused on the AR-5 backwater area. The contaminants of concern are DDT, DDE, DDD, benzene, ethylbenzene, chlorobenzene, 1,2-dichlorobenzene, 1,4-dichlorobenzene, and 1,2,4-trichlorobenzene. Such services shall be performed only after the U.S. Army and Pine Bluff Arsenal determine that they are legally permissible, justified, and authorized.

7.3.3 The contractor shall provide a report of the findings from subtasks 7.3.1 and 7.3.2. The report shall include tables and figures of findings, determination, if any, of Phillips Creek contamination to fish tissue, surface water and sediment quality in the Arkansas River, Phillips Creek confluence, downstream transects and AR-5 backwater area.

Option 7.1: Migration Rate of Phillips Creek Contamination

7.4.1 The contractor shall propose services to evaluate the migration rate of contaminated suspended sediment, bedload, surface water, and potential DNAPL from Phillips Creek to the Arkansas River, both upstream and downstream of the newly constructed diversion channel, as well as the diversion channel's effectiveness in reducing downstream contaminant transport in Phillips Creek. Such services shall be performed only after the U.S. Army and Pine Bluff Arsenal determine that they are legally permissible, justified, and authorized.

7.4.2 The contractor shall provide a report of the findings from subtask 7.4.1. The report shall include tables and figures of findings, analysis and a determination of the rate of migration of contaminated suspended sediment and bedload, surface water, and potential DNAPL in Phillips Creek to the Arkansas River upstream and downstream of the newly constructed diversion channel.

Task Number 7.4 – RCRA Maintenance

The contractor shall provide services to inspect and repair, if needed, the leachate collection system at the 83 Hazardous Waste Landfill. Inspection and repair are limited to above ground instrumentation, pumps and sensors. Frequency Once/year

Task Number 8 – Domestic Water Supply Well Removal

The contractor shall provide services to remove/abandon water wells on PBA property that are no longer used and/or no longer provide sufficient production of high-quality drinking water. All remaining drinking water wells are screened in the Sparta Aquifer and extend ~800ft – 1000ft. one per year. The following wells will be abandoned: Well #15, Well #16, Well #17, Well #18.

For well abandonment on government property in Arkansas, the contractor should comply with the current Arkansas Water Well Construction Commission (AWWCC) Rules and Regulations Governing Water Well Construction, Pump Installation, and Well Abandonment, along with applicable Arkansas statutes and licensing requirements. The work should be performed by an appropriately licensed contractor and should include well identification, removal of pumps and other equipment where practicable, placement of approved plugging materials, prevention of surface-water or cross-aquifer contamination, surface restoration, and submission of required abandonment documentation. Special conditions—such as flowing or artesian wells, contaminated wells, obstructed wells, or wells connecting multiple aquifers—should be coordinated with the AWWCC before work begins.[1]

Because the well is located on government property, the contractor must also comply with the applicable federal agency contract, project specifications, environmental requirements, and agency approvals. For U.S. Army or USACE projects, relevant Unified Facilities Guide Specifications, USACE engineering manuals, and project-specific environmental and safety plans may apply. USACE

EM 1110-2-1914, Design, Construction, and Maintenance of Relief Wells, should be consulted when abandoning relief wells or wells associated with dams, levees, seepage control, or foundation drainage. These documents supplement—but do not replace—the legally applicable Arkansas well-abandonment requirements.[2]

The contractor should also use appropriate technical guidance, including ASTM D5299, which provides guidance for decommissioning groundwater wells, monitoring devices, boreholes, and similar facilities. Depending on the well's construction and purpose, ASTM D5092, ASTM D5784, and AWWA A100 may provide additional information concerning well construction, annular seals, filter packs, grouting, and documentation.[3–6] If contaminated groundwater, soil, sediment, drilling fluids, purge water, or investigation-derived waste is present, the contractor must comply with applicable federal and Arkansas environmental requirements and obtain approval for handling, characterization, transportation, and disposal.

The contractor's compliance package should generally include the proposed abandonment method, well-construction records, license information, plugging-material specifications and quantities, daily abandonment logs, photographs, equipment-disposal records, surface-restoration documentation, surveyed well coordinates, and the required AWWCC and agency completion reports. The controlling requirements should be established from the applicable federal contract and permits, current AWWCC rules, site-specific regulatory requirements, and incorporated technical standards. ASTM standards and USACE manuals should not be treated as independently mandatory unless they are incorporated into the contract or otherwise directed by the Government.[1–6]

Task 9: Travel
The contractor will need to travel to the site to complete various aspects of the requirements listed in Task Number 001 through Task Number 008 above.

References

[1] Arkansas Water Well Construction Commission, *Rules and Regulations Governing Water Well Construction, Pump Installation, and Well Abandonment*; Arkansas Code Title 17, Chapter 50: https://www.arkleg.state.ar.us/ArkansasCode

[2] U.S. Army Corps of Engineers, *EM 1110-2-1914, Design, Construction, and Maintenance of Relief Wells*: https://www.publications.usace.army.mil

[3] ASTM International, *ASTM D5299, Standard Guide for Decommissioning of Groundwater Wells, Vadose Zone Monitoring Devices, Boreholes, and Other Devices for Environmental Activities*: https://www.astm.org

[4] ASTM International, *ASTM D5092, Standard Practice for Design and Installation of Groundwater Monitoring Wells*: https://www.astm.org

[5] ASTM International, *ASTM D5784, Standard Guide for Use of Hollow-Stem Augers for Geoenvironmental Exploration and Installation of Subsurface Water-Quality Monitoring Devices*: https://www.astm.org

[6] American Water Works Association, ***AWWA A100, Water Wells*****: https://www.awwa.org**